

Abstract

Many crustaceans rely on olfactory cues to navigate their aquatic environment. They detect odors using an array of chemosensory hairs located on their antennae. Through rapid flicking movements, these antennae draw in water samples and deliver chemical stimuli to olfactory receptors to “sniff” the water. This sensory mechanism allows crabs to locate food, identify mates, and detect predators. However, in recent years, seawater chemistry has changed due to increased pollution by microplastics, heavy metals, and other organic chemicals which can interfere with chemosensory. In this study, we investigate whether polluted seawater impairs chemosensing and predator detection in the crab *Hemigrapsus oregonensis*, and *Hemigrapsus nudus*. We found that crabs exposed to pollution treatments exhibited significantly lower metabolic rates and reduced antenna flicking frequency compared to those kept in regular seawater. These results indicate that pollution suppresses chemosensory behavior, likely diminishing the crabs' ability to detect and respond to predators. We hope our findings will provide more information about how pollution can affect chemosensory function across marine arthropods, and be used as a future resource to develop safer alternatives to current organic chemicals and pollutants to protect aquatic life.

Introduction

Hairy shore crabs (*Hemigrapsus oregonensis*) are native to Puget Sound and range from Resurrection Bay, Alaska, to Bahía de Todos Santos in Baja California. This species inhabits the intertidal zone, including open mudflats, algal mats, eelgrass beds, and exposed beaches. They possess small, flexible, bristle-like projections called setae that extend through the cuticle. Setae are primarily used to detect chemical cues and vibrations, helping crabs navigate, locate food (such as diatoms and green algae), and avoid predators. While setae are found on the legs, they are especially dense on the antennules. Crabs frequently perform a flicking motion with their antennules, drawing seawater over the setae and increasing the delivery rate of molecules to chemosensory receptors. This behavior enables crabs to distinguish the spatial distribution of odors, allowing them to track odor sources or detect predators.

However, under high pollution levels, chemical cues can be masked by heavy metals and organic pollutants. A common heavy metal in Puget Sound is copper, which originates from anti-fouling paints on boat hulls and brake pad dust from vehicles. Current acute and chronic water quality standards for copper are 4.8 µg/L and 3.1 µg/L, respectively. Previous studies have shown that copper exposure in crabs reduces hemolymph magnesium levels and leads to copper accumulation in the hepatopancreas, gills, carapace, heart, testes, and hypodermis.

Organic pollutants are persistent, carbon-based chemicals that accumulate in the environment and living organisms. Common organic pollutants in Puget Sound include tire chemicals, herbicides, and other compounds found in stormwater and runoff. Tire-related chemicals of concern include 6PPD, 6PPD-q, and tire and road wear particles (TRWPs). 6PPD and 6PPD-q are known to cause mortality in coho salmon during spawning. Herbicides such as 2,4-D and

atrazine are widely used in agricultural and residential settings to control weeds. Many pollutants go unnoticed and remain largely unregulated because toxicity thresholds vary widely due to their chemical diversity and differing environmental stability. Many of these pollutants are endocrine disruptors that can negatively affect the development and reproduction of wildlife. The combined effects of heavy metals and organic pollutants also pose risks to human health through contaminated drinking water and bioaccumulation in food webs.

Although previous studies have shown that increased plastic and copper pollution can impair the ability of crabs to detect or respond to predation cues, this pilot study specifically examines how Puget Sound pollutants affect chemosensing of predators by *Hemigrapsus oregonensis*.

Respiration rate and antenna flicking will be used as proxies for stress response to determine predator detection. Increased antennae flicking will indicate that crabs detect a predator, and this will be accompanied by increased oxygen consumption as a physiological stress response. We predict that pollutants will impair the chemosensory ability of shore crabs, causing them to fail to detect nearby predators and therefore show no stress response. Through this study, we aim to understand how chemosensing varies across regions with different pollutant profiles and how this may affect population dynamics between shore crabs and predators such as the invasive green crab, *Carcinus maenas*.

Methods

Habitat: Crabs were kept in clear plastic tanks at an ambient temperature of 24C and a salinity of 35 ppt. Tanks were continuously oxygenated and water was changed once per week.

Treatment groups: Two tanks were used: Tank 1 served as the control (no pollutants), and tank 2 served as the polluted treatment. During the first week of the experiment, the polluted tank contained the following contaminant to simulate environmental pollution: 500 ug/L herbicide, 6.849 grams of tire pellets, 500ug/L motor oil, and a 3x3 foil sprayed with copper paint to simulate leeching. During the second week of the experiment, herbicide and motor oil concentrations were reduced by half. Due to the change in pollution concentration, “high pollution” is referred as the polluted treatment during week two and “low pollution” is referred as the polluted treatment during week three. Tire pellets and the copper foil were reused.

Experimental groups: a total, four groups of crabs were used for the experiment. Each group consisted of two *H. oregonensis* and one *H. nudus*. Before the experiment, each crab was weighed and its sex was identified. Crabs were then placed into small glass jars. Jars labeled “A” indicated predator exposure while Jars labeled “B” indicated no predator exposure.

To prevent crabs from visually detecting predators through the glass, all jars were wrapped in a dark plastic tint. Although only two of the four groups were tested for predator detection, all jars were wrapped identically to maintain consistent conditions across all experimental groups. The jars were then placed into their corresponding treatment tanks (control or polluted) for the duration of the experiment.

In summary:

Jar 1A: no pollutants + no predator exposure (baseline)

Jar 1B: no pollutants + predator exposure

Jar 2A: pollutants + no predator exposure

Jar 2B: pollutants + predator exposure

Antenna flicking: Crabs that weren't assigned predator exposure (Jar 1A and 2A) were assessed for antenna flicking first. Their jars were removed from their corresponding tanks and placed under a phone camera, where they were recorded for one minute. Crabs assigned to predator exposure (Jar 1B, 2B) remained in their jars. A *C. maenas* crab was placed in their tanks for 15 minutes. Because the opening at the top of the jar was small, a small plastic pipette was used to collect water near the gills of the *C. maenas* and gently expelled into the jar. This method allowed the shore crabs to detect chemical cues (odors) from the predator crab, despite being physically isolated from it. After 15 minutes had passed, the jars were removed and placed under a phone camera where they were recorded for one minute. Each crab was then carefully examined for antenna flicking which is defined by the rapid, initial downstroke of the antennules.

Metabolic activity: a resazurin assay was used weekly as a proxy for stress. Resazurin is a non-toxic, cell permeable dye that fluoresces upon reduction during cellular respiration. The change in fluorescence over time serves as an indicator of relative respiration rate which is used as a proxy for stress. Shore crabs were placed in wellplate chambers containing 20 mL of resazurin working solution. Every 30 minutes, a 200 uL sample was transferred from each chamber into a 96-well plate. Samples were collected at three time points: T0, T1, T2. After the final collection, crabs were rinsed with salt water and returned to their original jars.

Data was recorded in Excel before being imported to R for statistical analysis. All figures were generated using ggplot. A one-way ANOVA was used to determine whether there were statistically significant differences in antenna flicking frequency and fluorescence among treatment groups and between high and low pollution conditions. A significance level of $\alpha=0.05$ was used, however, results were interpreted cautiously due to the small sample sizes, crab mortality, and the short experimental timeframe.

Results

Metabolic activity was highest in crabs exposed to predators without pollution. The control group (no pollution, no predator exposure) had the second highest metabolic activity. Crabs exposed to high pollution alone and crabs exposed to both pollution and predators showed similar metabolic activity, which was notably lower. Crabs exposed to low pollution alone had the third lowest metabolic activity, closely followed by crabs exposed to the combination of high pollution and predators.

Unlike metabolic activity, antennae flicking rates remained relatively constant across most treatments but varied between pollution levels. Crabs in the control group averaged the highest rate at 111 flicks per minute across both weeks. Crabs exposed to predators under high pollution averaged approximately 110 flicks per minute, as did crabs exposed to predators under

low pollution. Crabs exposed to low pollution alone averaged 106 flicks per minute. In contrast, crabs exposed to high pollution alone had the lowest flicking rate by a large margin, averaging only 34 flicks per minute.

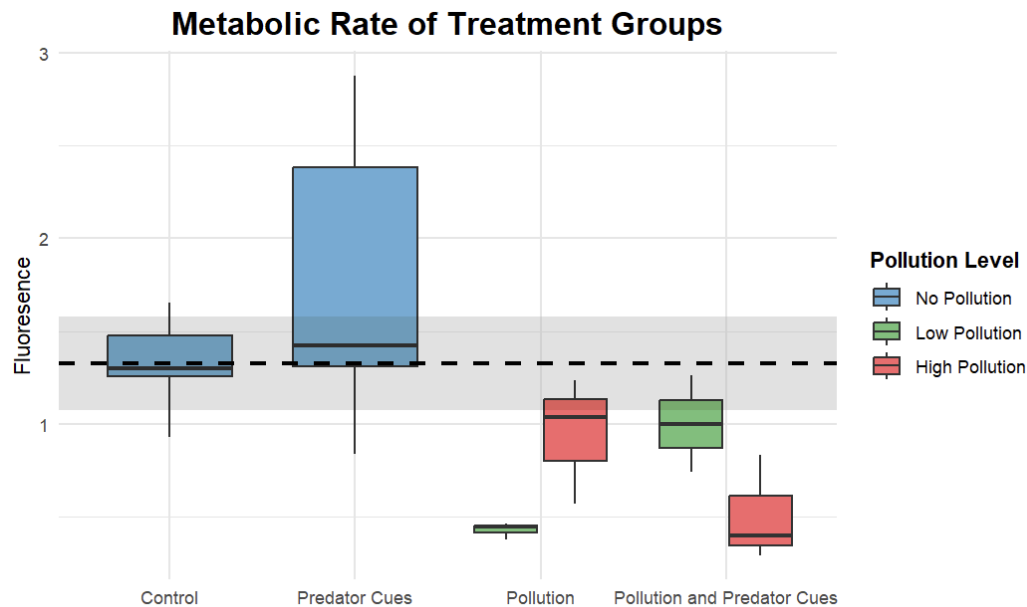


Figure 1. The final fluorescence (T2) in each of the four treatment groups after standardization. Groups are colored based on pollutant exposure level. Dashed line and highlighted region represents the mean control value and the standard deviation for more accurate comparison.

Mean Antennae Flicking by Treatment and Pollution Level

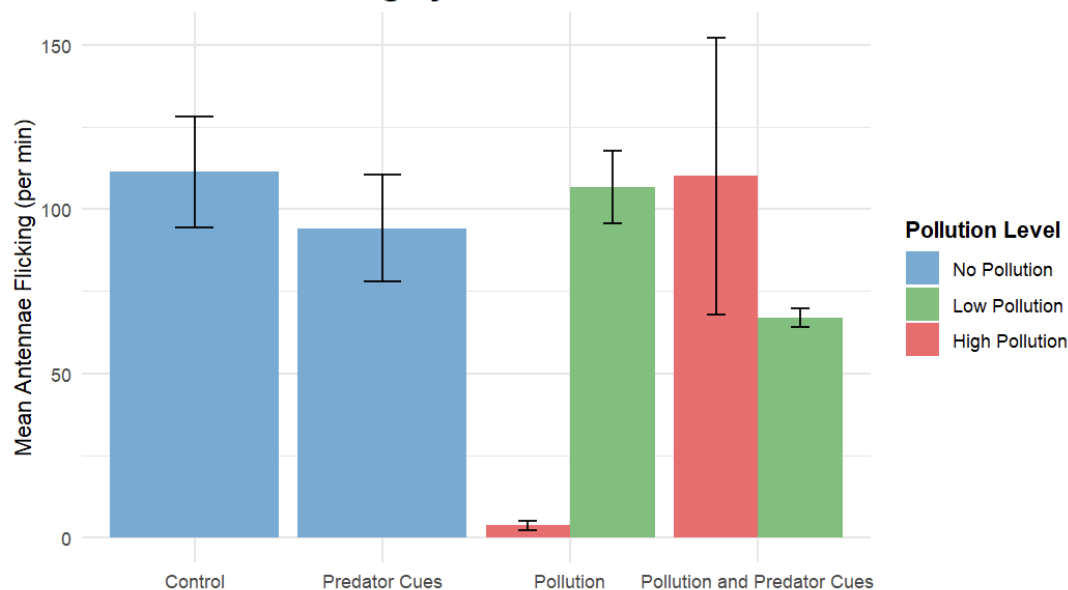


Figure 2 The mean antennae flicking per minute in each of the four treatment groups with the standard deviation. Groups are colored based on pollutant exposure level.

Discussion

Several limitations should be considered when interpreting our results. After the one-week acclimation period, all crabs in jar 2B and one *Hemigrapsus nudus* in jar 1B were found dead. This forced us to adjust the motor oil and herbicide concentrations and rapidly acclimate a new set of crabs to the polluted treatments, which may have introduced variability into our data. Additionally, handling and other experimental disturbances could have induced stress, potentially elevating metabolic rates. Furthermore, rapid antennae flicking does not necessarily indicate active processing of chemical cues, and manual counts of flicking are subject to human error.

Despite these caveats, our data indicate that pollution did affect the hairy shore crab (*Hemigrapsus oregonensis*). A one-way ANOVA on metabolic rates yielded a p-value of 0.0431, leading us to reject the null hypothesis. Crabs exposed to both predators and pollution exhibited lower metabolic rates, suggesting that they did not detect the presence of *Carcinus maenas* nearby. Similarly, antennae flicking rates in pollutant treatments were comparable to those of the control group, regardless of predator presence. This similarity leads us to conclude that crabs in the pollution treatments were unable to detect *C. maenas*. Notably, crabs exposed to high pollutant levels had the lowest flicking rate of any treatment, making them a clear outlier; we interpret this as an indication that these crabs largely ceased sampling their surrounding water.

Given that this was a pilot study with a limited experimental timeframe, it remains uncertain whether shore crabs would detect predator chemical cues after longer or repeated exposure. A related study on *Carcinus maenas* found that, rather than becoming desensitized, crabs became hypersensitive to sea bream odor over time, exhibiting stronger antipredator responses such as reduced foraging, increased hiding, and escape behaviors (Burnette et al., 2026). Importantly, chemosensory function in *C. maenas* was not impaired by pollutants, in contrast to the results of the present study. This discrepancy raises the following question for future research: Given sufficient time and repeated exposure, would *H. oregonensis* also become hypersensitive to predator odors, or would pollutant exposure prevent such sensitization? If shore crabs can develop sensitivity to predators, this could alter population dynamics given that *C. maenas* is a highly invasive species.

Beyond this unanswered question, it is clear that continued pollutant accumulation will have consequences, especially if no viable remediation solutions are implemented. Moreover, warmer temperatures and low pH have been shown to disrupt chemosensing in other crustaceans and even intensify the risks associated with plastic pollution (Ohnstad et al., 2025). These interacting stressors could have cascading effects on the food web, extending far beyond Puget Sound. Protecting chemosensory function in native species is therefore likely essential for maintaining resilient food webs and supporting the effective management of both native and invasive crab populations.

References:

- Burnett, J. W., Ohnstad, H., Jones, A. M., Hardege, J. D., & Bartels-Hardege, H. D. (2026). Olfactory Cues in the Odour Plume of Predatory Fish Reduce Foraging and Elicit Anti-Predator Behaviour in the European Green Crab *Carcinus maenas*. *Animals*, 16(5), 828. <https://doi.org/10.3390/ani16050828>
- Bjerregaard, P., & Vislie, T. (1986). Effect of copper on ion- and osmoregulation in the shore crab *Carcinus maenas*. *Marine Biology*, 91(1), 69–76. <https://doi.org/10.1007/BF00397572>
- Greer, J. B., Dalsky, E. M., Lane, R. F., & Hansen, J. D. (2023). Tire-Derived Transformation Product 6PPD-Quinone Induces Mortality and Transcriptionally Disrupts Vascular Permeability Pathways in Developing Coho Salmon. *Environmental Science & Technology*, 57(30), 10940–10950. <https://doi.org/10.1021/acs.est.3c01040>
- Krång, A.-S., & Ekerholm, M. (2006). Copper reduced mating behaviour in male shore crabs (*Carcinus maenas* (L.)). *Aquatic Toxicology*, 80(1), 60–69. <https://doi.org/10.1016/j.aquatox.2006.07.014>
- Ohnstad, H., Burnett, J., & Hardege, J. D. (2025). The Effects of Combined Stress from pH and Microplastic-Derived Odours on the European Green Crab *Carcinus maenas*'s Olfactory Behaviour. *Animals*, 15(4), 464. <https://doi.org/10.3390/ani15040464>
- Waldrop, L. D., & Koehl, M. A. R. (2016). Do terrestrial hermit crabs sniff? Air flow and odorant capture by flicking antennules. *Journal of The Royal Society Interface*, 13(114), 20150850. <https://doi.org/10.1098/rsif.2015.0850>